

Lab 3: Component Modelling & Architectural Pattern Selection

Student Startup Crowdfunding Platform (SSCP)

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Date: 16-09-2026

Architecture Selection

We chose Layered Architecture for the Student Startup Crowdfunding Platform (SSCP).

1. Architectural Choice

The SSCP is organised into three layers: a Presentation Layer (User Interface Component), a Business Layer (Campaign Manager, Milestone Approval, Payment Service and Notification Components), and a Data Layer (Database Component). Each layer only communicates with the layer directly below it through well-defined interfaces, which keeps the design close to the model demonstrated for the Coffee Kiosk example in this lab.

2. Two Reasons for the Choice

- Clear separation of concerns: The system has three fairly distinct responsibilities — presenting campaigns to founders/backers, running the campaign-and-milestone business logic, and persisting records/handling payment transactions. A layered structure maps naturally onto these responsibilities and matches how the requirements were already grouped in the Lab 2 backlog (campaign creation, browsing, investing, progress tracking).
- Manageable scale for the project: SSCP is a single, moderate-sized student platform with one primary workflow (create campaign → pledge → milestone approval → fund release), not a system that needs independently-scaled services. A Layered Architecture is easier to build, test and demonstrate within the lab timeframe than a Microservices split, while still keeping the codebase modular through the component interfaces shown in the diagram.

3. Security Advantage

Because the Presentation Layer can only reach the Data Layer through the Business Layer's components, sensitive operations such as fund release and license/payment validation are never exposed directly to the user interface. The Payment Service Component is the sole component permitted to talk to the payment gateway and the Database Component, so campaign and pledge data cannot be read or modified except through the Milestone Approval and Campaign Manager interfaces — reducing the attack surface for unauthorized access to financial data.

4. Performance Benefit

Grouping the Campaign Manager, Milestone Approval, Notification and Payment Service Components in one Business Layer means related operations (e.g. approving a milestone and immediately releasing funds) can be executed as in-process calls rather than network calls between separately-deployed services, avoiding the network latency that a Microservices approach would introduce for this size of application.

Component Summary

- User Interface Component — Presentation Layer
- Campaign Manager Component — Business Layer
- Milestone Approval Component — Business Layer
- Payment Service Component — Business Layer
- Notification Component — Business Layer
- Database Component — Data Layer